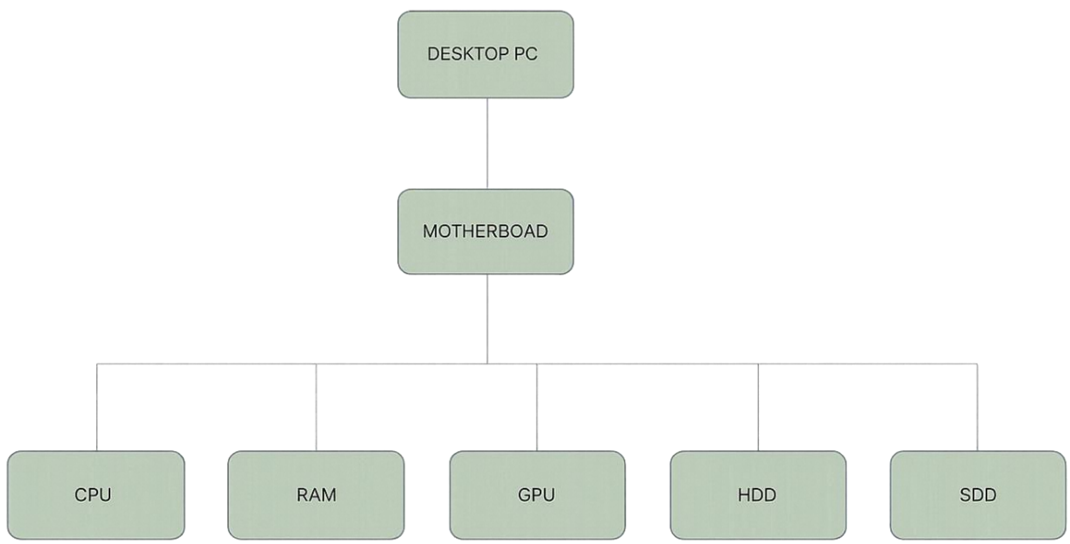
Day 1 :- Overview of Hardware Components Theory : Introduction to motherboards, CPUs, Ram , storage (HDD, SSD). Importance of each component in a computer system :-

Ans:-

Tasks 1 :- Draw a labeled diagram of a desktop PC, clearly marking the motherboard, CPU, RAM, GPU, and both HDD and SSD storage components.



**Tasks 2 :- Create a comparison table showing the main differences between HDD and SSD storage in terms of speed, durability, and typical use cases.

Hint: Think about which one is faster,**

which is more shock-resistant, and where you would usually find each type (laptop, gaming PC, etc.).

Feature	HDD(Hard Disk Drive)	SSD(Solid State Drive)
Speed	Slower	Much Faster
Storage Method	Storage Method	Storage Method
Boot Time	Longer	Very Quick
Durability	Less durable due to moving parts	More durable, no moving parts
Noise	Produces noise	Silent
Power Consumption	Higher	Lower
Cost per GB	Cheaper	More expensive
Noise	Produces noise	Silent
Typical Use Cases	Large file storage, budget PCs	Gaming PCs, laptops, operating systems
Shock Resistance	Low	High

Hint Answer:

- SSD is faster than HDD.
- SSD is more shock-resistant because it has no moving parts.
- HDD is commonly used for large storage, while SSD is preferred in laptops and gaming PCs for better performance.

Taks 3:- Write a short explanation (2-3 sentences each) describing the role of the CPU, RAM, and GPU in running a game like PUBG or Free Fire on a PC.

- **CPU (Central Processing Unit) :-**

  The CPU is the brain of the computer. In games like PUBG or Free Fire, it processes player inputs, game logic, AI behavior, and physics calculations. A faster CPU helps the game run more smoothly and reduces lag.

- **RAM (Random Access Memory) :-**

  RAM temporarily stores game data that the computer needs to access quickly. It helps load maps, textures, and game files while playing. More RAM allows smoother gameplay and better multitasking without slowing down the system.

- **GPU (Graphics Processing Unit) :-**

  The GPU is responsible for rendering images, videos, and 3D graphics in the game. It processes visual effects, shadows, textures, and animations to provide high-quality graphics. A powerful GPU improves frame rates and makes gameplay look smoother and more realistic.

Tasks 4 :- Imagine you want to upgrade your PC to run the latest version of FIFA smoothly. List which hardware components you would consider upgrading and explain why for each.

GPU (Graphics Card) :-

To improve graphics quality, frame rate, and support modern visual effects.

CPU (Processor) :-

To handle game calculations, AI, and physics efficiently.

RAM :-

To ensure smooth gameplay and reduce stuttering. 16 GB is recommended for modern games.

SSD Storage :-

To reduce game loading times and improve system responsiveness.

Power Supply (PSU) :-

A stronger PSU may be needed if a more powerful GPU or CPU is installed.

Cooling System :-

Better cooling prevents overheating during long gaming sessions.